

2025 秋个人研究计划 (Research Proposal)

重要提示:

- 不要修改或删除已有内容或文本格式包括字体、字号、字色以及各类间距边距等;
- 正文部分英文: 300-400 词、中文: 450-600 字;
- 写作完毕后导出为 PDF 格式, 用班级+姓名格式命名文件。

基本信息 (Information)

班级 (Class): B5

姓名 (Name): 陈悦

学号 (Student ID): 20251800834

所在小组编号 (Group No.): 14

研究计划标题 (Proposed Research Title):

中文: 《2004-2024 年内蒙古典型露天煤矿区“生态-地下水”动态响应机制的机器学习模拟与验证》

英文: Machine Learning Simulation and Validation of the Dynamic Response Mechanism of "Ecology-Groundwater" in Typical Open-pit Coal Mining Areas of Inner Mongolia from 2004 to 2024

第一段: 研究背景 (Introduction / Background)

写作指引: 此段对应大纲的第1部分。请介绍宏观背景, 强调话题重要性, 并引出核心文献及其贡献, 最后自然地过渡到现有研究的局限性。

In arid and semi-arid regions such as Inner Mongolia, China, the groundwater system constitutes a critical lifeline for ecological security and sustainable development, with the coupled 'eco-groundwater' system being particularly fragile. However, in the paper 'Varying effects of...', empirically revealed that mining activities, especially open-pit mining, are disrupting this delicate balance with unprecedented intensity, creating ecological and hydrological effects that exhibit significant spatial heterogeneity. While this study successfully maps the complex picture of mining impacts, it also exposes limitations in our current understanding of the underlying driving mechanisms and our ability to achieve accurate prediction.

第二段: 问题陈述与研究空白 (Problem Statement / Research Gap)

写作指引: 此段对应大纲的第2部分。清晰地、有理有据地论证现有研究的不足 (即“研究空白”), 并强调填补此空白的必要性和价值。

A central research gap is that existing methods struggle to quantify the dynamic response of the 'eco-groundwater' system under different mining scenarios and lack a robust assessment model that can integrate physical mechanisms with observational data while maintaining high generalizability. Therefore, precisely deciphering this dynamic response mechanism and developing a reliable simulation tool is not

only crucial for advancing the theory of human-land system coupling in mining areas but also represents a pressing practical need for precise environmental monitoring, remediation, and risk early warning.

第三段：研究方案与预期贡献 (Research Plan & Expected Contribution)

写作指引：此段浓缩大纲的第3、4、5部分。用简洁、有力的语言，依次完成三件事：1. 正式提出你的研究问题/目标；2. 简要介绍你将采用的研究方法；3. 预告你研究的预期成果与意义。

This research aims to quantitatively characterize the dynamic "ecology-groundwater" response in typical Inner Mongolian open-pit coal mining areas from 2004 to 2024, under a shallow fissure-pore aquifer context. Through an integrated methodology that couples remote sensing ecological indices, historical mining data, and physics-constrained machine learning models, we will simulate and validate the spatiotemporal interactions within this critical system. Ultimately, the study is expected to yield a validated model that reveals key drivers and thresholds, providing a scientific foundation for sustainable groundwater management and ecological restoration in these arid, ecologically vulnerable mining regions.